Written HW 9 — Full Solutions

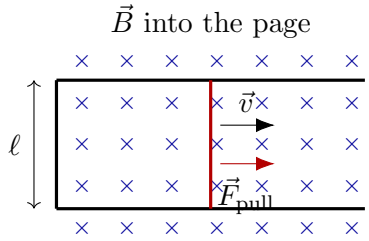
Problem statements from Knight, *Physics for Scientists and Engineers* (5th ed., Global Edition), Chapter 30. $e = 1.60 \times 10^{-19}$ C, $m_p = 1.67 \times 10^{-27}$ kg throughout.

30.2 (3 pts)

A 10-cm-long wire is pulled along a U-shaped conducting rail in a perpendicular magnetic field. The total resistance of the wire and rail is 0.20 Ω. Pulling the wire at a steady speed of 4.0 m/s causes 4.0 W of power to be dissipated in the circuit. (a) How big is the pulling force? (b) What is the strength of the magnetic field?

MODEL: At a *steady* speed the wire's net force is zero, so the pulling force exactly balances the magnetic drag force. That means every joule of work the puller does shows up as dissipated electrical power: $P = F_{\text{pull}} v$. That alone gives part (a). For part (b), the motional emf is $\varepsilon = B\ell v$, which drives a current $I = \varepsilon/R$ and dissipates $P = \varepsilon^2/R = (B\ell v)^2/R$ — solve that for B.

VISUALIZE:



The U-shaped rail and sliding wire form a closed loop; the wire of length $\ell = 10$ cm is dragged to the right at constant v, so the induced current I through resistance R dissipates power P exactly as fast as the pulling force supplies it.

SOLVE: (a)

$$P = F_{\text{pull}} v \;\Rightarrow\; F_{\text{pull}} = \frac{P}{v} = \frac{4.0 \text{ W}}{4.0 \text{ m/s}} = \boxed{1.0 \text{ N}}.$$

(b)

$$P = \frac{(B\ell v)^2}{R} \;\Rightarrow\; B = \frac{\sqrt{PR}}{\ell v} = \frac{\sqrt{(4.0 \text{ W})(0.20 \ \Omega)}}{(0.10 \text{ m})(4.0 \text{ m/s})} = \frac{0.894}{0.40} = \boxed{2.2 \text{ T}}.$$

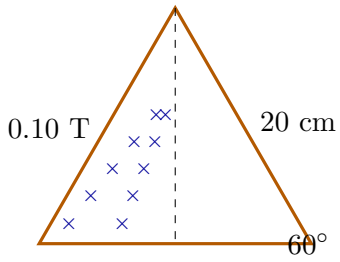
REVIEW: Check the two parts against each other: with $B = 2.2$ T, the current is $I = \varepsilon/R = B\ell v/R = (2.2)(0.10)(4.0)/0.20 = 4.5$ A, and $F_{\text{pull}} = BI\ell = (2.2)(4.5)(0.10) = 1.0$ N — exactly the part (a) answer, as it must be, since both routes describe the same steady-state force balance. 2.2 T is a strong field (comparable to an MRI magnet), which makes sense: dissipating 4.0 W through a fairly brisk 4.0 m/s pull with only 0.10 m of wire needs a hefty field to generate much current.

30.11 (2 pts)

The metal equilateral triangle in **Figure EX30.11**, 20 cm on each side, is halfway into a 0.10 T magnetic field. (a) What is the magnitude of the magnetic flux through the triangle? (b) If the magnetic field strength decreases, what is the direction of the induced current in the triangle?

MODEL: Flux only accumulates where the field actually penetrates the loop: here the field exists only in the *left* half of the triangle (split by its own vertical symmetry axis) — the right half sits in a field-free region and contributes nothing. So the relevant area for $\Phi = BA$ is just that left half-triangle, a 30–60–90 right triangle with base 10 cm (half of the 20 cm side) and height equal to the full altitude of the equilateral triangle, $20 \sin 60^\circ$ cm.

VISUALIZE:



The dashed line is the triangle's vertical symmetry axis; only the half marked with ×'s (left of the dashed line) lies in the field — the right half is field-free. That half-triangle has base 0.10 m and height $(0.20 \text{ m}) \sin 60^\circ = 0.1732$ m.

SOLVE: (a)

$$A_{\text{half}} = \tfrac{1}{2}(\text{base})(\text{height}) = \tfrac{1}{2}(0.10 \text{ m})(0.1732 \text{ m}) = 8.66 \times 10^{-3} \text{ m}^2$$

$$\Phi = BA_{\text{half}} = (0.10 \text{ T})(8.66 \times 10^{-3} \text{ m}^2) = \boxed{8.7 \times 10^{-4} \text{ Wb}}.$$

(b) The flux into the loop is decreasing. By Lenz's law the induced current opposes that decrease, so its own field must point *into* the loop (the same direction as the original field) to try to prop the flux back up. By the right-hand rule, that requires the induced current to flow $\boxed{\text{clockwise}}$.

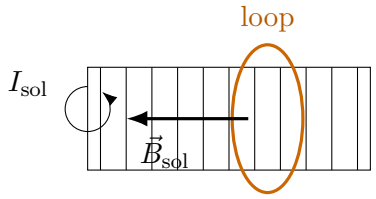
REVIEW: Sanity check on the area: the full equilateral triangle has area $\frac{\sqrt{3}}{4}(0.20 \text{ m})^2 = 1.73 \times 10^{-2}$ m^2, and half of that is 8.66×10^{-3} m^2 — matches. The more important point is conceptual: even though the whole triangle is one conducting loop, only the half that's actually immersed in $\vec{B}$ contributes to Φ — $\Phi = \int \vec{B} \cdot d\vec{A}$ is zero wherever $\vec{B}$ is zero, loop or no loop.

30.12 (3 pts)

The current in the solenoid of **Figure EX30.12** is increasing. The solenoid is surrounded by a conducting loop. Is there a current in the loop? If so, is the loop current cw or ccw?

MODEL: A long solenoid's field is (nearly) confined to its interior, so a loop encircling the solenoid — even though its own area is a bit larger than the solenoid's cross section — still captures essentially the full internal flux, since almost no field lines exist outside to *not* be captured. As I_{sol} grows, that enclosed flux grows, and Lenz's law says the loop's induced current opposes the increase.

VISUALIZE:



(viewed from the left, looking right along the axis)

As shown, I_{sol} circulates counterclockwise when viewed from the left end looking right along the axis, which by the right-hand rule drives $\vec{B}_{\text{sol}}$ pointing *left* (toward that same viewer) inside the solenoid.

SOLVE: Since I_{sol} is increasing, $\vec{B}_{\text{sol}}$ (pointing left) is growing stronger, so the leftward flux through the surrounding loop is increasing. By Lenz's law the loop's induced current must create a field that opposes that increase — i.e., a field pointing *right* inside the loop. Applying the right-hand rule to a rightward-pointing field (same left-looking-right viewpoint) requires the current to circulate clockwise.

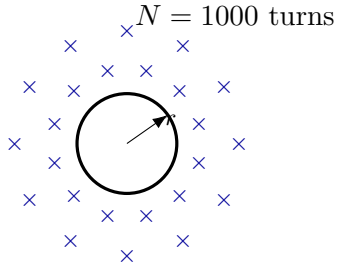
REVIEW: Yes, there is an induced current, and its direction is opposite the sense of I_{sol} itself (ccw in the solenoid, cw in the loop) — makes sense, since the loop is actively fighting against the solenoid's growing field. Notice the loop's exact radius never entered the reasoning: as long as it's larger than the solenoid and coaxial with it, it encloses the same flux a snug-fitting loop would, since the field just outside a long solenoid is essentially zero.

30.16 (3 pts)

A 1000-turn coil of wire 1.0 cm in diameter is in a magnetic field that increases from 0.10 T to 0.30 T in 10 ms. The axis of the coil is parallel to the field. What is the emf of the coil?

MODEL: With the coil's axis parallel to $\vec{B}$, the flux through each turn is simply $\Phi = BA$ (no $\cos\theta$ factor needed). Faraday's law for an N-turn coil gives $\varepsilon = N\left|\dfrac{d\Phi}{dt}\right| = NA\left|\dfrac{\Delta B}{\Delta t}\right|$, since only B — not the coil's area or orientation — is changing.

VISUALIZE:



The coil's diameter is 1.0 cm, so $r = 0.50$ cm $= 0.0050$ m.

SOLVE:

$$A = \pi r^2 = \pi(0.0050\text{ m})^2 = 7.85 \times 10^{-5}\text{ m}^2$$

$$\left|\frac{\Delta B}{\Delta t}\right| = \frac{0.30\text{ T} - 0.10\text{ T}}{10 \times 10^{-3}\text{ s}} = 20\text{ T/s}$$

$$\varepsilon = NA\left|\frac{\Delta B}{\Delta t}\right| = (1000)(7.85 \times 10^{-5}\text{ m}^2)(20\text{ T/s}) = \boxed{1.6\text{ V}}.$$

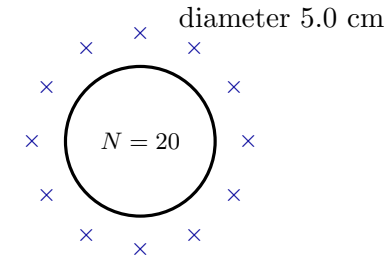
REVIEW: A single turn in the same tiny (1.0-cm-diameter) coil would only produce 1.6 mV — it's entirely the $N = 1000$ multiplier that turns a modest field ramp into a usable 1.6 V. Also notice the sign of ΔB never entered the calculation; Faraday's law only cares about the *magnitude* of the rate of change for the emf's size (its direction is a separate, Lenz's-law question).

30.17 (4 pts)

CALC A 5.0-cm-diameter coil has 20 turns and a resistance of 0.50 Ω. A magnetic field perpendicular to the coil is $B = 0.020t + 0.010t^2$, where B is in tesla and t is in seconds. (a) Find an expression for the induced current $I(t)$ as a function of time. (b) Evaluate I at $t = 5$ s and $t = 10$ s.

MODEL: The field is perpendicular to the coil (so $\Phi = BA$ again) but now changes with time according to a given function, so $\varepsilon(t) = NA\left|\dfrac{dB}{dt}\right|$ must be found by differentiating $B(t)$ first. Ohm's law then converts that emf into a current, $I(t) = \varepsilon(t)/R$.

VISUALIZE:



$B(t) = 0.020t + 0.010t^2$ T
$R = 0.50\ \Omega$

The coil's radius is 2.5 cm = 0.025 m, so $A = \pi(0.025\text{ m})^2 = 1.96 \times 10^{-3}\text{ m}^2$.

SOLVE: (a)

$$\frac{dB}{dt} = 0.020 + 0.020t \quad (\text{T/s})$$

$$\varepsilon(t) = NA\left|\frac{dB}{dt}\right| = (20)(1.96 \times 10^{-3}\text{ m}^2)(0.020 + 0.020t) = 7.85 \times 10^{-4}(1 + t)\text{ V}$$

$$I(t) = \frac{\varepsilon(t)}{R} = \frac{7.85 \times 10^{-4}(1 + t)\text{ V}}{0.50\ \Omega} = \boxed{1.6 \times 10^{-3}(1 + t)\text{ A}}.$$

(b)

$$I(5\text{ s}) = (1.57 \times 10^{-3})(1 + 5) = \boxed{9.4 \times 10^{-3}\text{ A}}$$

$$I(10\text{ s}) = (1.57 \times 10^{-3})(1 + 10) = \boxed{1.7 \times 10^{-2}\text{ A}}.$$

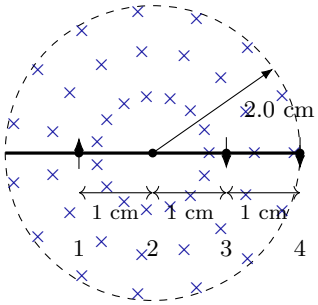
REVIEW: $I(t)$ comes out perfectly linear in t, which makes sense: dB/dt itself is linear in t (since $B(t)$ has a t^2 term), and everything else in $\varepsilon(t) = NA|dB/dt|$ is a constant. Check the ratio: $I(10)/I(5) = 11/6 = 1.83$, and indeed $(1 + 10)/(1 + 5) = 11/6$ — consistent. Even though the field $B(t)$ itself is always increasing, it's really the *rate of change* dB/dt that sets the current, and that rate keeps climbing too (the field's growth is accelerating), which is why I keeps climbing at $t = 10$ s rather than leveling off.

30.18 (3 pts)

INT The magnetic field in **Figure EX30.18** is decreasing at the rate 0.10 T/s. What is the acceleration (magnitude and direction) of a proton initially at rest at points 1 to 4?

MODEL: A changing B induces a circulating electric field even with no wire present (Section 30.6). Inside a circular field region of radius R, that induced field has magnitude $E(r) = \dfrac{r}{2}\left|\dfrac{dB}{dt}\right|$ at radial distance r from the center, directed tangentially; the sense (cw/ccw) comes from Lenz's law, as if a loop *were* there. A proton at rest then simply obeys Newton's second law with this electric force: $\vec{a} = e\vec{E}/m_p$, in the same direction as $\vec{E}$ (positive charge).

VISUALIZE:



$\vec{B}$ points into the page and is decreasing, so by Lenz's law the induced $\vec{E}$ circulates *clockwise* (to drive a current that would try to maintain the inward flux). Points 1–4 lie 1 cm apart on a line through the center (point 2); clockwise circulation points *up* on the left side (points 1) and *down* on the right side (points 3, 4), as drawn.

SOLVE:

$$a(r) = \frac{eE(r)}{m_p} = \frac{er}{2m_p}\left|\frac{dB}{dt}\right|$$

Point 1 ($r = 1.0$ cm $= 0.010$ m, up):

$$a_1 = \frac{(1.60\times10^{-19}\text{ C})(0.010\text{ m})(0.10\text{ T/s})}{2(1.67\times10^{-27}\text{ kg})} = \boxed{4.8\times10^{4}\text{ m/s}^2\text{, up}}.$$

Point 2 ($r = 0$, the center):

$$\boxed{a_2 = 0}.$$

Point 3 ($r = 1.0$ cm, down): same magnitude as point 1, opposite side,

$$a_3 = \boxed{4.8\times10^{4}\text{ m/s}^2\text{, down}}.$$

Point 4 ($r = 2.0$ cm $= 0.020$ m, down): twice the radius of point 3,

$$a_4 = \boxed{9.6\times10^{4}\text{ m/s}^2\text{, down}}.$$

REVIEW: $a(r)$ is exactly linear in r — doubling the radius (point 3 to point 4) exactly doubles the acceleration. This is the same "uniform source spread over a disk" scaling that shows up for $\vec{E}$ inside a

uniformly charged sphere or $\vec{B}$ inside a uniform current-carrying wire: the induced field grows linearly out from the center, then (outside the field region, not asked here) would fall off as $1/r$ instead. Point 2 sits exactly on the axis of symmetry, where the tangential contributions from all sides cancel — a proton dropped there, unlike anywhere else inside the field region, would feel nothing at all.